

Activity Guide - Encryption Design Challenge



Your challenge: Design a brand-new encryption method to protect messages. Your system should scramble a message so that only someone with the right rule, pattern, or key can decode it. It must balance security and usability. Get creative — think letters, numbers, symbols, colors, sounds, or even physical gestures!

Step One - The Message

Pick a simple **message** you want to protect and write it below:

Step Two - Encryption Method or Key

Describe your encryption **method** and **key** in the space below:

Use a sentence starter, if necessary:

"To decode my message, you need to..."

"The key is a _____ that tells you how to..."

"Each symbol means a _____, and the pattern is..."

Illustrate or diagram how your rule would work on **letter**, a **number**, and a **special character**:

Step Three - Encrypt Your Message

Encrypt your original message using your encryption method:

Step Four - Method Instructions for Decryption

Write a set of instructions for another group to use in order to **decrypt** your message:

Step Five - Swap Activity Guides with Another Group to Test Methods

Use the space below to follow the instructions in Step Four to decrypt the message in Step Three:

Give feedback once you decrypt the message:

- How easy was it to decrypt?
- How strong was it? Could someone decode it without knowing the key?